EE1100 – Basic Electrical Engineering
Tutorial 6: Operational Amplifiers

Q1. The input and output signals of an amplifier are shown in Fig. 1. Find the voltage gain of the amplifier. Is it a linear amplifier? Explain your answer. Design the OP-AMP circuit with appropriate values of circuit components.

[Voltage gain = 2000; linear]

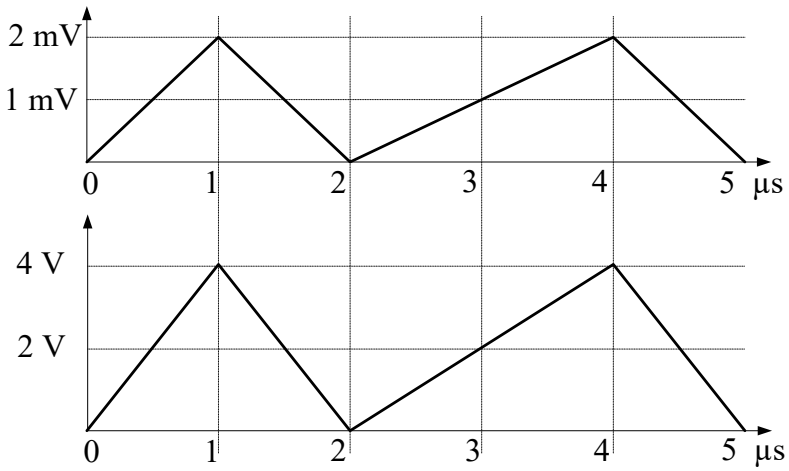


Fig. 1

Q2. The OP-AMP shown in Fig. 2 has an input voltage V_i of 1mV. Calculate (a) the voltage gain and (b) the output voltage.

[Gain = -80; V_0 = -80mV]

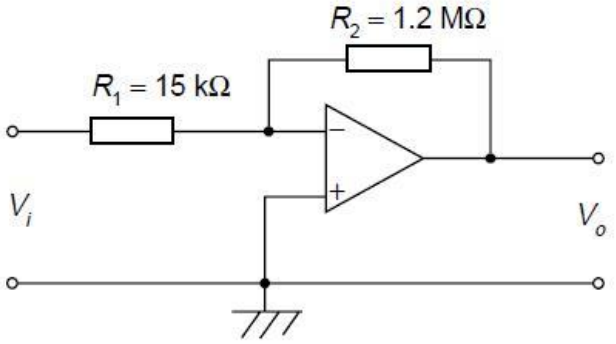


Fig. 2

Q3. If the input voltage for the circuit shown in Fig. 3 is -0.5V, determine (a) the voltage gain (b) the output voltage.

[Gain = 3.2058; output voltage=-1.6029V]

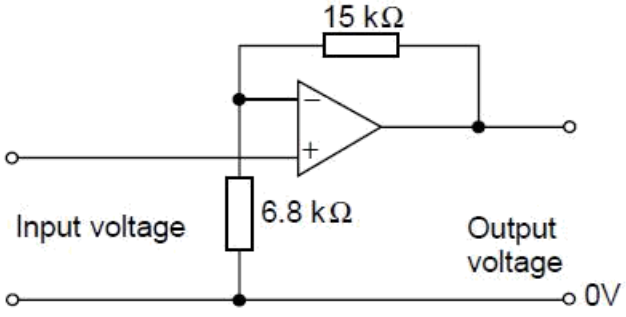


Fig. 3

Q4. For the circuit shown in Fig. 4, determine the output voltage V_0.

[V_0 = -3.9V]

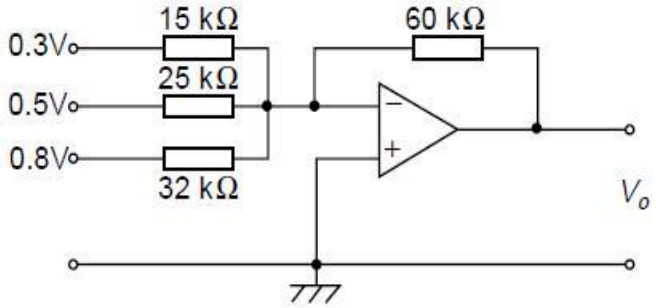


Fig. 4

Q5. In the arrangement shown in Fig. 5, determine the output voltage, V_0, if:

(a) V_1 = 4mV and V_2 = 0 (b) V_1 = 0 and V_2 = 6mV (c) V_1 = 40mV and V_2 = 30mV
(d) V_1 = 25m Vand V_2 = 40mV

[(a) -60mV (b) 90 mV (c) -150 mV (d) 225 mV]

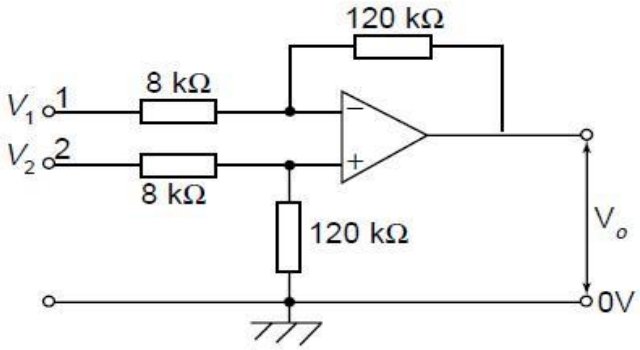


Fig. 5

Q6. A steady voltage V_i of 0.75 V is applied to the circuit in Fig. 6, having component values of R = 200 kΩ and C=2.5 μF. Assuming that the initial capacitor charge is zero, determine the value of the output voltage 100 ms after applying the input.

[V_0 = -0.15V]

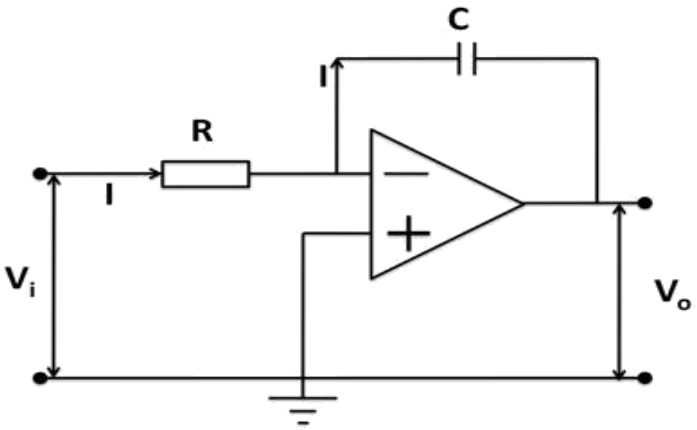


Fig. 6

Q7. In the circuit shown in Fig. 7, R_s = 1 kΩ and R_L = 10 kΩ. For the OP-AMP, A = 10^5, R_i = 100 kΩ and R_0 = 100 Ω. For V_0 = 10V, calculate V_s and V_0/V_s .

[V_S = 10.0001V; V_0/V_S = 0.9999]

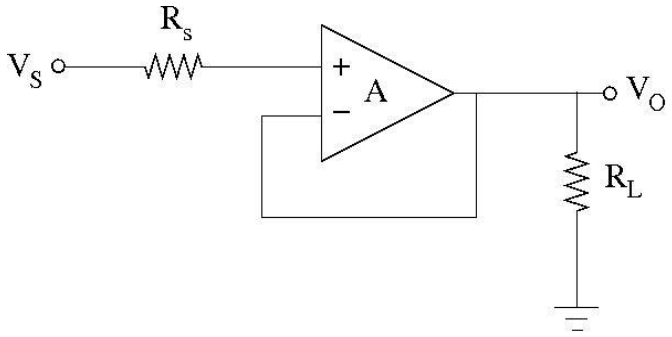


Fig. 7

Q8. The photo diode has sensitivity of 10 μA/mW of incident radiation. If V_0 = 8 V, calculate input radiation power to photo diode?

[W = 6 Watt]

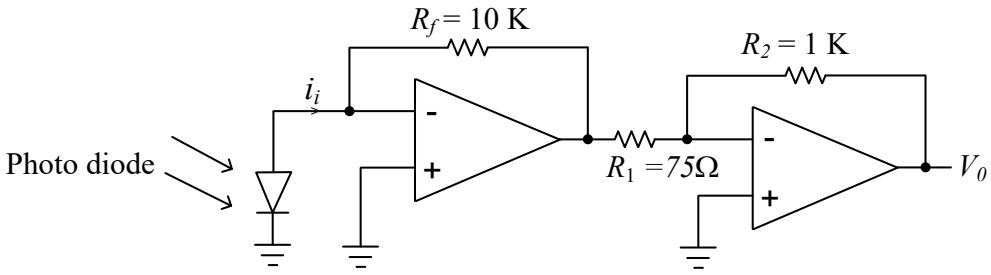


Fig. 8

Q9. Find V_0 and i_0 in the circuit in Fig. 9

[V_0 = -2 V; i_0 = -1 mA]

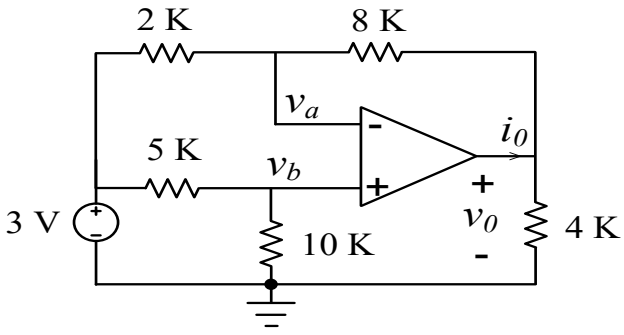


Fig. 9

Q10. Find the condition for the resistances to get $v_0 = v_2 - v_1$ for OP-AMP circuit shown in Fig. 10

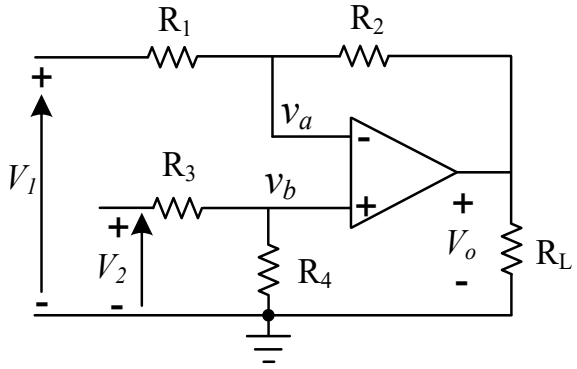


Fig. 10

Q11. Suppose a physical system described by $z = 2x-3y$, where x and y are voltages corresponding some inputs in system and within threshold value of OP-AMP. Draw an OP-AMP circuit that makes z as output variable. Confine to standard resistor values.

Q12. An historic bridge is showing signs of deterioration. Until renovations can be performed, it is decided that only cars weighing less than 1600 kg will be allowed across. To monitor this, a four-pad weighing system is designed. There are four independent voltage signals, one from each wheel pad, with 1 mV = 1 kg. Design a circuit to provide a positive voltage signal to be displayed on a DMM (digital multimeter) that represents the total weight of a vehicle, such that 1 mV = 1 kg. You may assume there is no need to buffer the wheel-pad voltage signals.